INDUSTRIAL PHARMACY - IST

B PHARMACY 5TH SEMESTER

2 MARKS NICHOD SOLUTIONS

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DESCRIPTION

- This video contains short questions of all units (unit 1 to Unit-5)
- Long important Questions + Syllabus → completed with pdfs
- These question helps in :-
  - directly 5-6 Questions
  - 2-3 short question (one long question)
  - Overview + Quick revision of all units
  - Industrial Pharmacy - I का complete निचोड़

MAHA-RIVISION

Q. Define Industrial Pharmacy

⇒ It is the study of formulation, manufacturing, quality control, and large-scale production of pharmaceutical products.

Eg. Production of tablet (granulation, compression, coating). Capsule, ointments, syrup → large scale batch production.

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UNIT-1

① Define Preformulation. Mention its two major goals.

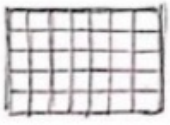
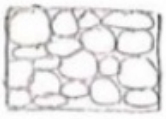
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Preformulation is the systematic study of physiochemical properties of a drug substance before formulation.
The aim is to collect essential information that will guide the design, formulation and development of a safe, stable and effective dosage form.

Eg. Before making tablet of Ibuprofen, its solubility, pKa, and flow properties must be studied.

Goals - To develop safe, stable and effective dosage form.
- To determine drug - excipient compatibility before formulation.

② Differentiate Crystalline Vs Amorphous forms.

Crystalline form	Amorphous form.
• The molecules arranged in a regular, repeating pattern. (like grid).	• The molecules arranged in a random or disordered way, like glass.
• Usually less soluble 	• Usually more soluble. 
• More stable (chemically & physically)	• Less stable (can degrade faster).
• Sharp Melting point.	• No sharp Melting point (soften over range)
Example - Paracetamol (stable).	Example - Amorphous insulin

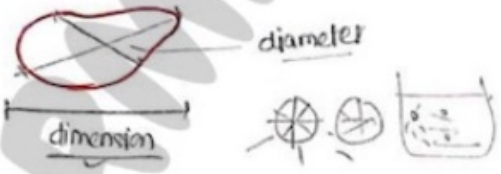
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③ Define Particle size & its significance in formulation.

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Particle size refers the diameter or dimension of a single particle or group of particles.

Eg. Particle size ↓ ⟶ Solubility ↑.



Significance -

- Affects dissolution rate (smaller particle ⟶ (faster) dissolution).
- Affects bioavailability and uniform mixing.

④ Define Angle of repose, Carr's index, Hausner ratio and their importance.

⇒

Angle of repose - It is the maximum angle formed between the surface of a (pile) of powder and the horizontal plane, when the powder is allowed to flow freely to form a heap.

$$\tan\theta = \frac{h}{r}$$

$$\theta = \tan^{-1} h/r$$

where, h = height of pile, r = Radius, θ = angle of repose.



Importance -

- Indicates (flow) property.

(Carr's index) - It is an indication of powder flowability based on the difference between (tapped) and bulk densities.

$$CI = \frac{\text{Tapped Density} - \text{Bulk density}}{\text{Tapped density (TD)}} \times 100$$

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Angle of Repose

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flow Properties	Angle of Repose
Excellent	less than 25
Good	25 – 35
fair (aid not needed)	36 - 40
Passable (may hang up)	41 - 45
Poor (must agitate/vibrate)	46 - 55
very poor	56 - 65
Very - very poor	>66

Carr's Index & Hausner's Ratio

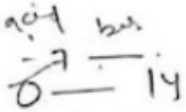
flow Property	Carr's Index	Hausner's Ratio
Excellent	≤ 10	1.00 – 1.11
Good	≤ 11-15	1.12 - 1.18
fair	16-20	1.19 - 1.25
Passable	21 - 25	1.26 - 1.34
Poor	26 - 31	1.35 - 1.45
Very Poor	32 - 37	1.46 - 1.59
Very Very poor	>38	>1.60

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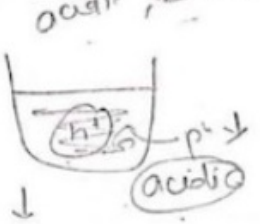
Hausner Ratio - It is the ratio of (Tapped) Density to Bulk Density and it reflects the inter-particle friction.

Hausner Ratio = Tapped Density / Bulk Density



⑤ Define (pH), pKa & significance of pKa.

⇒



pH - Power of (Hydrogen) [OR] Potential of (Hydrogen). ↓

pH is the negative logarithum of the hydrogen ion concentration in a solution.

$$pH = -\log [H^+]$$

Eg. Solubility of (aspirin) can be enhanced by addition of alkaline buffer.

pKa - (Ionization constant) - It is the (pH) at which (50%) of the drug molecules are ionized and 50% are unionized.

$$pKa = -\log_{10} Ka.$$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ (Ionized form)
(Unionized)

Eg. Weak acid absorb better in stomach (lower pH).

$$pKa = \frac{(H^+)(CH_3COO^-)}{(CH_3COOH)}$$

Significance -

- Helps predict (ionization) and solubility of drug at (GI ph.)

⑥ Define Partition coefficient, give (two) applications.

⇒

The ratio of concentration of a drug in (oil) phase (lipid) to its conc^n in (aqueous) phase (water) when the two are in equilibrium.

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- usually measured in octanol - water system.

$$K_{olw} = \frac{X_o}{X_w}$$

where, X_o = conc. of drug in organic (oily phase)

X_w = conc^n. of drug in aqueous phase.

>1 lipo
<1 hydro

Application -

- Predict membrane permeability → important in drug absorption.

⑦ Define polymorphism and its formulation significance.

⇒

- It is the ability of a solid material (drug or excipients) to exist in two or more crystalline forms with different arrangement of molecules in the crystal lattice but having the same chemical composition.

eg. Ritonavir had formulation failure due to unexpected polymorph.

Significance -

- Different forms show different solubility, Stability
- Melting point.



⑧ Define hydrolysis, oxidation & one method to prevent each.

⇒

Hydrolysis - The decomposition of drug molecules due to reaction with water and get hydrolysed (breakdown of drug substances).

- Mostly drugs with function group ester, amide, lactam undergoes hydrolysis

eg. Aspirin, Atropine etc... $CH_3COOCH_3 + H_2O \xrightarrow{H^+} CH_3COOH + CH_3OH$

Prevention - Use anhydrous solvent or dry packaging.

- Use of buffer to maintain optimum pH.

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Oxidation - Degradation by reaction with oxygen (loss of electrons or addition of oxygen). often involves free radical chain reactions.

- It occurs due to oxygen, Heavy metals ion, Light.

Eg. Rancidity oil.

$$2Mg + O_2 \longrightarrow \overset{+2\ \ -2}{2MgO}$$

Oxidizing agents

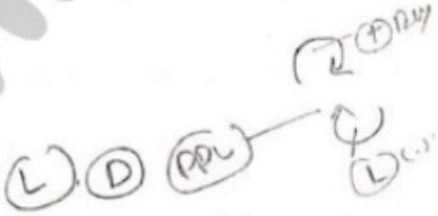


Prevention -

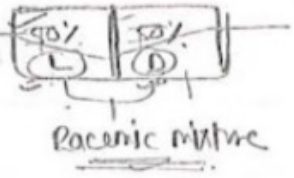
- Use antioxidants (ascorbic acid, BHT).
- Replace air with inert gases (nitrogen, CO_2).

⑨ Define racemisation & polymerization.

⇒



Racemisation - It is on conversion of an optically active drug (enantiomer) into its mirror image (opposite enantiomers).

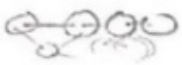


- This may reduce or completely change drug activity.

Eg. L-Epinephrine → D-Epinephrine.

Polymerization - It is a chemical reaction where two or more drug molecules combine to form high molecular weight products.

- Often leads to loss of potency and sometimes toxicity.



Eg. Formaldehyde → Paraformaldehyde (degradation).

⑩ What is BCS classification? Mention two advantages.

⇒

- Biopharmaceutics classification system, based on solubility and Permeability. It helps in understanding drug dissolution, absorption and bioavailability which are critical for oral dosage forms.

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	Solubility (S)	Permeability (P)	Example
Class I	High (h)	High (h)	Paracetamol
Class II	Low (lo)	High (h)	Phenytoin
Class III	High (h)	Low (lo)	Ranitidine
Class IV	Low (lo)	Low (lo)	Furosemide

Advantages -

- Helps predict oral absorption.
- Useful in biowaivers (Class I drugs may not need BA/BE studies).
  - regulatory provision allowing drug companies to skip costly, time-consuming in vivo (human/animal) bioequivalence (BE) studies.

⑪ Mention any two preformulation consideration important for parenterals.

⇒

- Solubility in suitable parenteral vehicle (water for injection). WFI
- Isotonicity requirement (0.9 % NaCl equivalent).

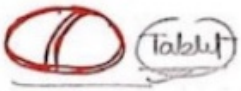
Eg. Injection solutions must match osmotic pressure of blood.

Others :-

- pH and buffering (pH → 3-9 usually acceptable).
- Must be stable during storage and sterilization
- Lyophilization (freeze-drying) used for unstable drugs.

UNIT-2 ↓

① Define tablet. Write two ideal Characteristics.



⇒

- A tablet is a compressed solid dosage form containing one or more medicaments, with or without excipients.
- It is usually circular, oval or oblong in shape and easy to manufacture, stable and convenient for patient.

Eg. Paracetamol tablet 500 mg.

Ideal characteristics -

- Should be free from defects (cracks, chips, discoloration).
- Should be uniform in weight & drug content.
- Must have sufficient hardness and mechanical strength.

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② Classify tablets (any two classes).

⇒

1. Tablet ingested Orally.
   - Compression tablet
   - Multiple compressed tablet
   - Sustained tablet.
2. Tablet used in Oral cavity.
   - Buccal tablet
   - Sublingual tablet
   - Dental cone.





③ Name any two tablet excipients with their uses.

⇒

Binders - It helps to bind powder particles together to form granules, improving mechanical strength of tablet. These agents are present in both dry or in wet form.

eg. Acacia, Gelatin etc...

Uses - Provide cohesion
- Prevent tablet breakage.



Disintegrants - These are facilitate breakup of tablet into smaller particles after swallowing → enhances dissolution and absorption.

eg. Sodium starch glycolate, Bentonite etc...

Uses - Helps tablet break apart.
- Ensure rapid disintegration and drug release.



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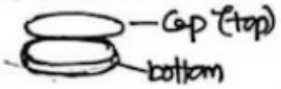
④ Write steps of wet granulation. dry gr

- Use binder solution in wet form. The steps required for this:-

(i) Mixing of API & excipients.
(ii) Preparing a dump mass
(iii) Screening - Granulation
(iv) Drying
(v) Drying - screening
(vi) Lubrication
(vii) Compression.



⑤ Give tablet compression problems.



Capping - The partially or completely separation of the top or bottom crown of the tablet from the main body of a tablet.

Lamination - The separation of a tablet into two or more distinct layers is called lamination.



Reason - Excess moisture, air entrapment, improper binder amount.

⑥ Mention two evaluation tests of tablets.

(i) Dissolution Test - It is the process by which a solute dissolve into a solvent or solution.

- The main purpose is to measure rate and extent of drug release from tablet.

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(ii) Disintegration Test - The time required to disintegrate the tablet is called disintegration time.

- It is mainly used to check bioavailability & therapeutic efficacy.

⑦ Define Bloom Strength & its significance.

⇒

- It is the measure of gel strength of gelatin, indicating its molecular weight and hardness.



Eg. Pharmaceutical gelatin bloom strength - 150 - 300 Bloom.

Significance -

- Higher bloom strength = stronger gelatin = better capsule shell quality.

⑧ Give Types of tablet coating.



⇒

- Tablet coating is the process of applying a thin, uniform and continous layer of coating material on the surface of tablets.
- The process is usually performed in a coating pan or fluidized bed coater.

Types of Coating.

(i) Sugar coating
(ii) Film coating.
(iii) Enteric coating.



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⑨ Mention two coating defects.

⇒

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- In this, we study about during coating, processing errors may lead to several defects in appearance or functionality of tablets.



(i) Blistering - The high temperature occurs during drying process cause rapid evaporation of solvent and produce an un-smooth coated surface with number of uneven spots are called blisters.

(ii) Chipping - It occurs when the film becomes dented and chipped from the edges of the tablet.



- The excess drying make the coat brittle.

⑯ Give Two advantages of film coating.

⇒

- Film coating is a modern tablet coating technique in which a thin, uniform layer of polymer film is applied to the surface of a tablet.

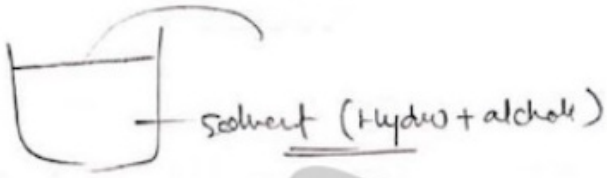


Advantages -

- Protect drug from moisture, light, oxygen.
- Mask bitter taste
- Increase mechanical strength.

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⑪ Define elixer & syrup.



⇒

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Elixers are clear, sweetned, hydroalcoholic oral liquid preparation intended for oral use, containing one or more active medicinal agents.

- Good in taste especially for children. eg. Paracetamol elixirs etc
- Absorbed more quickly in the gastrointestinal tract.

Syrup is a sweet, viscous, concentrated aqueous solution that contains sugar (usually sucrose). with or without medicinal substances.

- 66.7% w/v sugar I.P. eg. Cough syrup.

⑫ Give evaluation of liquid orals.

⇒

(i) p^H Measurement - Ensure stability, solubility and palatability.

- Antitussives / Expectorants → slightly acidic.
- Measured using calibrated p^H meter.

(ii) Microbial limit Test - Must be free from pathogenic organisms like E. coli, salmonella, Pseudomonas.

- Total viable count should be within specified pharmacopoeial limits.

⑬ Define Suspension & Emulsion with example.



⇒

Suspension. is a biphasic liquid dosage form of medicament in which the finely divided solid particles are suspended or dispersed in a liquid with the help of suspending agents.

eg. Chalk mixture, Paracetamol suspension etc.

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Emulsion. is an biphasic liquid dosage form in which two immiscible liquids are mixed together to form a stable emulsion with hlp of emulsifying agent.

Eg. Vanishing cream.



(14) Two packaging consideration for liquid orals.

⇒

- Should be packed in air-tight, light-resistant containers.
- Must have child-resistant closures.
- Adequate headspace to avoid spoilage.



UNIT-3 CAPSULES + PELLETS

(1) Define hard gelatin capsule, mention two raw materials.

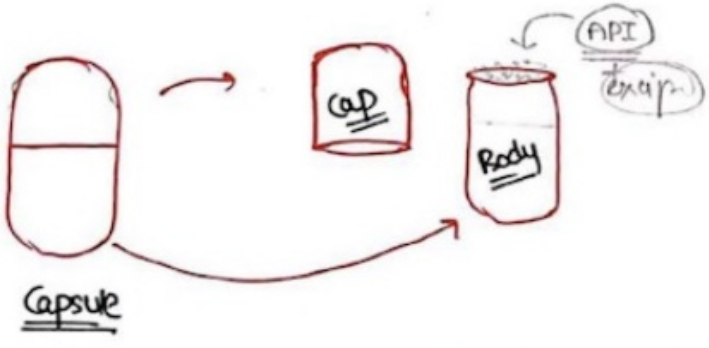
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- These are two piece solid dosage forms made of a gelatin shell that consists of a "body" and a "cap" designed mainly for filling dry materials like powders, granules, pellet and small tablet.

Eg. Amoxicillin capsule.

Raw materials -

- Gelatin (main component)
- Water (plasticizer)
- Colorant.



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② What is capsule size ? Give one example.



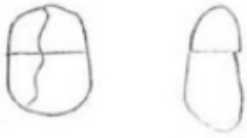
⇒

- Capsules are solid unit dosage forms in which a drug or mixture of drug is enclosed inside a small, easy to swallow shell.
- Capsule size number indicates the volume capacity of empty capsule shells.
  - Common adult size – 0 ~ 0.68 ml
  - Moderate fill size – 1 ~ 0.50 ml
  - High Dose size – 00 ~ 0.95 ml

000 → largest size
00, 0, 1, 2, 3, 4, 5 smallest

Eg. Paracetamol 500 mg → size 0 or 00 capsule.

③ Mention two defects of hard gelatin capsule.



⇒

- Cracks or splits – Small cracks develop on the body or cap when gelatin becomes too dry or brittle.
- Improper Cap-Body fit – The cap does not lock properly on the body due to wrong trimming, size mismatch, or deformation.

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④ Mention any two capsule evaluation tests.

⇒

- Weight variation – Individual capsules are weight to confirm that each unit contains almost the same total weight and does not show excessive variation.



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Disintegrating Test - It checks the time required for the capsule shell to break and release its contents under standards conditions, ensuring proper breakdown in the digestive system.



⑤ Define BAF & m/g factor.

⇒



Base Adsorption factor.

- It is the number of grams of liquid base required to produce a uniform fill mass with 1 gram of drug.

BAF = Weight of base absorbed x Weigh of drug.

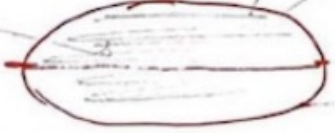
M/g factor - It indicates the amount of formulation (in mg) that can be filled per 1 g of mixture.

⑥ Two advantages of soft gelatin capsules.

⇒

- Soft gelatin capsules are one piece, sealed capsules made from soft, elastic gelatin, designed mainly for filling liquids, oils and semi-solid drug formulation.

Eg. Vitamin E softgel.



Advantages -

- Suitable for oily or low-melting drugs.
- Provide better bioavailability due to liquid / semi solid fill.
- Easy to swallow and elegant appearance.

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⑦ Two evaluation tests for soft gel capsules.

⇒

(i) Leakage / Seal Integrity Test - This test checks whether soft gelatin capsules are properly sealed and ensures that no liquid leaks out of the capsule.



(ii) Assay - correct drug quantity.

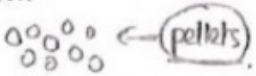
(iii) Disintegration test - must disintegrate properly.

⑧ Give two storage conditions for soft gels.

⇒

- Store at low humidity (20-30%)
- Temperature below 25°C
- Protect from light (gelatin may degrade).
- Airtight bottles prevent moisture exchange.

⑨ Define pellets & write two formulation requirement.



⇒

- Pellets are small, free-flowing, (spherical) semi-spherical solid units containing drug and and excipients, usually with a size between 0.5 - 1.5 mm. They are prepared by a process called pelletization.



eg. Pantoprazole enteric coated.

Formulation requirement.

- should have good hardness & spherical shape
- should have uniform size distribution.
- Good flow propnty for capsule / tablet filling.

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⑩ Define extrusion - spheronization (pelletization process).

⇒

- Extrusion is the wet mass is pased through an extrude, where it is forced through small cylindrical holes to produce long spaghetti like strands of uniform thickness.
- Spheronization is the wet extruder are broken and rolled in a fast rotating spheronizer plate, which rounds them into smooth, spherical pellets.



⑪ Give one evaluation test of pellets.

⇒

(i) Sphericity test - Determines roundness of pellets. (value close to 1.0).

(ii) Friability test - Pellets should not break during handling

(iii) Size distribution - Uniform size is essential.



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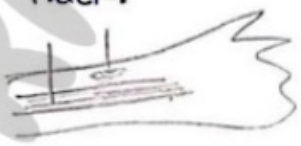
UNIT-4 → PARENTERAL & OPHTHALAMIC PREPARATIONS

① Define parenterals & mention two advantages.

⇒

- These are sterile (free from any kind of microbes) preparation intended to be administered by injection through skin or mucous membranes into body tissues. These products bypass the gastrointestinal tract.

Eg. IV, IM injections.



Advantages.

- Produce rapid therapeutic action (immediate effect).
- Useful when patient is unconscious or unable to take orally.
- 100 % bioavailability.

② Define LVP and SVP with volume range.



⇒

- Large Volume Parenterals (LVP) → These are sterile solutions intended for intravenous fusion use in volume more than 100 mL and without any antimicrobial preservatives. Volume ranges → >100ml (250mL, 500 & 1000 mL)

Eg. Dextrose 5%, Normal saline 0.9%, Ringer's lactate etc.

- Small Volume Parenterals (SVPs) → These are sterile preparation for injection, infusion, or implantation with volume of 100 mL or less. Range → ≤100mL (1mL, 2mL, 5mL, 10 mL, 30mL, 50mL)

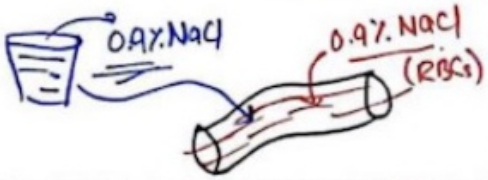
Eg. Atropine injection, vit. B12 injection etc

③ Define isotonicity & its pharmaceutical importance.

⇒

- Isotonicity ensures that the osmotic pressure of the parenteral product matches that of body fluids.

Eg. 0.9% nacl is isotonic with blood.



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Importance.

- Prevents hemolysis (cell bursting) or crenation (cell shrinkage).
- Essential for eye drops & injection to avoid irritation → ensures safety.

④ Name any two additives used in injections.

⇒

- Antioxidants - Sodium metabisulfite
- Preservatives - Benzyl alcohol
- Buffer - Citrate, Phosphate buffer
- Tonicity adjusters - NaCl, Dextrose.

⑤ Mention any two vehicles used for injections.

⇒

- Water for Injection (WFI) - commonly used, pyrogen-free water used in preparation of injection.
- Aqueous buffer - for pH adjustment
- Non-aqueous solvents - eg. Propylene glycol, Sesame oil.

⑥ Write two key steps of aseptic processing.

⇒

- It is Highly clean zone used for critical operations like filling and sealing eg. Vial filling room.
- Sterilization of equipment, container & closures.
- Sterile filtration of solution (0.22 μm filter).
- Filling under Laminar Air flow (LAf).

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⑦ Define lyophilization & give one advantage.

⇒

- Lyophilization also called freeze drying, It is a process of removing water by freezing the product and then removing ice by sublimation under vaccum.

Eg. Vaccines, antibiotics like Penicillin G.



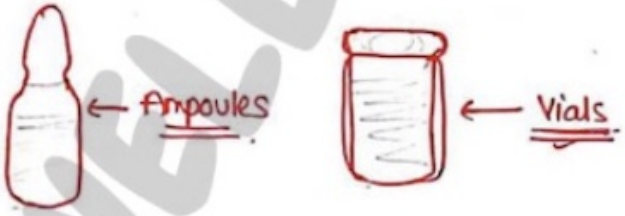
Advantages -

- Improves stability of heat sensitive drugs.
- Allows easy transport and storage.

⑧ Name two containers for injections, write one point on sealing.

⇒

- Ampoules
- Vials.



Sealing point - Ampoules are sealed by melting the glass neck, while vials are sealed with rubber closures and aluminum crimps.

Eg. Dextrose 5% is packed in glass bottles / vials.

⑨ Mention two QC tests for parenterals (sterility, clarity).

⇒

(i) Sterility Test - Ensures product is free from microorganisms. The test is performed by incubating the product in sterile culture media for 14 days and checking for microbial growth.

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(ii) Clarity Test - It ensures that the injection is clear, colourless and free from visible particles.

(iii) Leak Test - This test ensures that the container is sealed properly and no contamination can enter.



⑦ What are pyrogens? Mention one method of pyrogen testing?

⇒

- Pyrogens are fever - causing substance, mainly endotoxins from Gram - negative bacteria.
- Even if a product is sterile, pyrogens may still be present, so this test confirm that pyrogen level is safe.

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(i) Bacterial Endotoxin Test (LAL Test) Modern Method -

Uses blood extract from horseshoe crab (Limulus Amebocyte Lysate)

- If endotoxin is present, the reagent forms a gel or turbidity.
- Very sensitive and fast.

⑧ Define Opthalmic Preparation with example.

⇒

- These are sterile medicines used in the eye. They must be free from particles, non - irritating and safe for the sensitive tissues of the eye.

Eg. Eye drops, Eye ointments.

Advantages -

- Give quick relief in local eye problems.
- Cause minimal systemic side effects.

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⑨ Define eye drops, list (two) formulation condiderations.

⇒

- These are clear, sterile liquid preparations meant for instillation into the eye.

Eg. Ciprofloxacin eye drops.

Formulation consideration.

- Must be isotonic (same osmotic pressure as tears)
- Must be sterile & pyrogen free.
- Use of appropriate buffer to maintain ph.

⑩ Mention (two) evaluation tests for ophthalmic preparations.

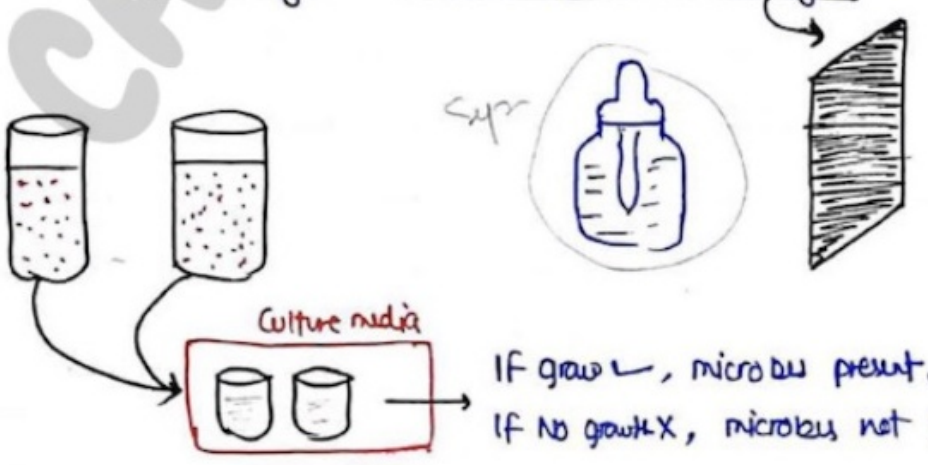
⇒

(i) Sterility Test - Ensures the product is free from microorganisms

- Done by membrane filtration or direct inoculation.
- Incubated for 14 days — no growth.

(ii) Clarity Test - Ensure solution is clear and particle-free.

- Observed against black & white background.



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UNIT-5 → COSMETICS, PACKAGING & AEROSOLS

① Define lipstick. Mention one key ingredients.



⇒

- Lipstick are solid sticks applied on lips to give color, shine and protection. They are one of the most popular makeup products used daily.

Key ingredients.

- Waxes (Beeswax, carnauba wax) - give shape & hardness.
- Oils (Castor oil, Lanolin) - Smoothness and shine.

② What is vanishing cream?



⇒ O/W

- These are oil in water emulsions that disappear quickly after application due to high water content. They are non-greasy, light and mainly used as foundation base or for oily skin.
  eg. Stearic acid - based vanishing cream.
- Used as base for fairness / skin care creams.
- Skin smoothing

③ Define SPF & its importance.

⇒

- Sunscreen are topical preparations that protect skin from harmful UV rays, preventing sunburn, tanning and premature aging. They are classified by SPF (Sun Protection factor).

SPF 50+

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- Higher SPF - better sun protection.
- Non-comedogenic and dermatologically tested.

④ Name two types of hair dyes.

⇒

- Hair dyes are cosmetic products used to colour hair temporarily or permanently. They may be chemical (oxidative) or herbal (Henna). Permanently dyes involve oxidation reactions.

Eg. Natural dye - Henna
Permanent dye - Ammonia based.

(i) Temporary Hair dyes
(ii) Permanent Hair dyes.

⑤ Define Aerosols & its advantages.

⇒

- Aerosols are pressurized pharmaceutical dosage forms in which the drug is dispensed as fine spray or mist using a pressurized container & valve system.

Eg. Inhaler for asthma, Nasal sprays etc...

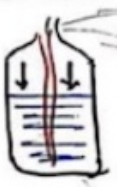
Advantages -

- Easy to use and portable
- Rapid onset of action
- Minimizes contamination.



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⑥ Define two types of propellants.



⇒

- These are gases or liquid that create pressure inside the aerosols container and help expel the drug as spray or foam.

Types - These are of two types:-

(i) Liquefied Gas Propellants - Store as liquid under pressure, vapourize on release.

Eg. Hydrocarbons - Butane, propane.

(ii) Compressed Gas Propellants - Gases filled under high pressure, not liquefied.

Eg. Co_2, Nitrogen.

⑦ Mention two parts of Aerosol valve.



⇒

- Valve is the control unit of the aerosol system. It is mechanical part that controls the release of drug from the aerosol container.

Parts of valve.

- Actuator - release product in spray form.
- Valve stem - passage for product.
- Dip Tube - drows product from bottom.

⑧ One evaluation Test for aerosols.

(i) Leak Test - This test is used to check that aerosol container is completely sealed or not. If there is any small leak, the aerosol will lose pressure.

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(ii) Spray pattern Test - This test is used to check that how the spray looks when it comes out of the nozzle.

⑨ Define Type I and Type III glass.

⇒

- Glass is considered as gold standard for packaging because it is non-reactive, transparent and provides excellent protection against environment factors.

Types of Glass.

(i) Type-I - Borosilicate Glass - Higly resistant to chemical / thermal shock
  - Used for injections, LVP, SVP. (eg. Ampoules)

(ii) Type III - Regular Soda-lime Glass - Moderate chemical resistant.
  - Used for oral liquid, tablets. (eg. Syrup bottles).

⑩ Give one QC test for glass containers.

⇒

(i) Light Transmission Test - Determines how much light passes through the container.
  - Used for amber bottles to ensure UV protection.

(ii) Hydrolytic Resistance Test - Purpose is to measure how much alkali (Na^+, K^+) leaches from glass into water.
  - Leached alkali can degrade the drug or change pH.

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⑪ Give (Two) factors influencing selection of (Containers).

⇒

- Chemical compatibility with drug.
- Protection from light, moisture, oxygen.
- Strength & rigidity.
- Cost and regulatory requirements.

⑫ Name (two) plastics used in pharmaceutical packaging.

⇒

- Plastics are synthetic polymers used extensively in packaging due to their (low) cost, versatility and resistance to breakage.

(i) Polyethylene (PE) - It is flexible, moisture-resistant, chemically inert. (eg) HDPE, LDPE bottles.

(ii) Polypropylene (PP) - It is high (heat) resistant, good chemical resistant, light weight.

(iii) Polyvinyl chloride (PVC) - It is Transparent, strong, good barrier to oils.

⑬ One (QC) test of rubber (closures).





⇒

(i) Penetrability Test - Ensures needle can pierce rubber stopper with minimum force.

(ii) Compressibility Test - Ensures closures properly after withdrawal of needle.

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① What is Hygroscopicity ? Give one example.

⇒

- It is the property of a substance to absorb moisture from the atmosphere.
- Hygroscopicity drugs lose stability, become sticky and show poor flow.

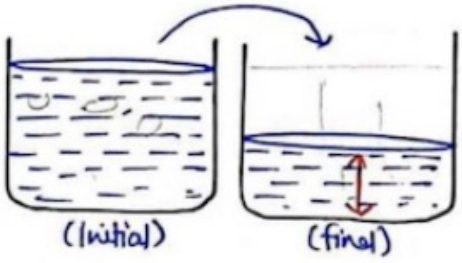
Eg. Sodium chloride, Calcium chloride.

② What is Sedimentation Volume ?

⇒

- It is the ratio of final (settled) volume of suspension to the original volume.

$$F = \frac{V_u}{V_o} \Rightarrow \frac{\text{final volume}}{\text{Initial volume}}$$



Significance -

- Higher F = better stability.

Eg. well-flocculated suspension has large sedimentation volume.

f = 1 → Good suspension

Questions for You :-

1. Define Industrial Pharmacy
2. Diff. b/w Disintegration & Dissolution
3. two superdisintegrant (Croscarmellose Sodium) used in tablets
4. Enteric Coated tablets
5. two plasticizers used in soft gelatin
6. important requirement of Rubber
7. two tonicity adjusting agents
8. pore size of sterile membrane filter

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